

**Team Tendon's
Project**

**Sustained Loop for
Effective Plastic Waste
Management and
Community Benefits**

Theme of idea

Solid, Liquid & Hazardous Waste Management.

Sub Theme

Plastic Waste & Circular Economy

Problem Title

Improper management of urban plastic waste causes severe air, water, and soil pollution, accelerates climate change, and critically threatens the health of humans, birds, and animals.

Problem statement & Background

Kerala has launched many new plastic waste management projects, but they still face major problems. Right now, people only take part to avoid paying heavy fines, not because they want a cleaner environment. They are very unhappy because local authorities charge a fixed monthly fee even when waste collection is irregular and unreliable. Because of this lack of trust, some people secretly burn or dump their waste outdoors.

Kerala generates about 3900 tons per day from 93 ULBs of which nearly 70% is wet waste. There is a large capacity gap for processing the waste within the city.

Existing plants report issues of odor and leachate which cause air water and soil pollution. And are objected by the neighbourhood. The capacity deficit leads to illegal dumping and increase in stray dog, snakes and rodent population. The solid waste dumping is one of the leading cause for blockage of drains, groundwater pollution and lake pollution.

None of the cities in Kerala have been able to manage all the waste. The facilities are inadequate and many projects are yet to take off.

Kerala has adapted a decentralized waste management policy which needs to be complimented by few centralized facilities. Real time data of actual solid waste generation/processed and disposed is not available in the cities. Centralized facilities require large land and must conform to siting guidelines by PCB. ULBs face financial constraints in operation and maintenance.

Who is affected by this problem

The entire society including Citizens, Farmers, Students, Industry, Govt Systems, Ragpickers, Waste workers and other organisms like Animals and birds are also affected by this problem.

Root cause of the problem?

- 1.Lack of proper effective projects to gain public support.
- 2.Lack of responsibility in individuals regarding one's own waste.
- 3.Negative attitude of public towards existing proper waste disposal projects.
- 4.Lack of data.
- 5.Poor community acceptance.
- 6.Lack of job description and organization of plastic waste management.
- 7.Lack of suitable land paparcel

Existing solutions and their limitations

Even though Kerala Government has initiated so many projects for proper plastic waste mangement,they face so many limitations.

Government plastic waste projects face heavy public backlash because residents suffer a financial loss by handing valuable recyclables—like bottles and cardboard—to official collectors for free, whereas local scrap shops pay cash for them.

Small groups producing minimal waste are frustrated regarding the constant user fee they have to pay as they are generating lower waste materials than those

generating high and all of them has to give the same amount.

local authorities rely heavily on strict administrative penalties and fines to force compliance. Thereby making the public to be a part of the initiatives just to escape from those arrears leaving the responsibility and civic sense aside. Consequently, bachelors and small households generating minimal waste bypass the system entirely—selling profitable items to scrap shops and illegally dumping the rest—which completely undermines the core purpose of these municipal green policies.

Kerala's waste management initiatives face deep public friction primarily because residents experience a glaring financial loss: instead of earning valuable cash from local scrap dealers for recyclable items like clean plastic bottles, metal cans, and cardboard, they must hand these materials over for free while simultaneously paying a mandatory fixed monthly fee to government collectors like the Haritha Karma Sena.

Thrust on decentralised solutions, handing over to piggeries, handover to HKS. The decentralised solutions are often not working. there are no qualified technical agencies, this leads to people being averse to operate household system. Community solutions are not objected

to and hence ULBs are hesitant in allocating land to the project. Current large capacity project require large extent of land

Expected outcome of solving this problem?

1. Plastic pollution can be reduced.
2. Strategy for gaining community acceptance.
3. Negative attitude of public can be turned into positive for proper waste management
4. Accumulation of plastic waste can be reduced.
5. Proper Plastic waste management obtained with public support.
6. No more plastic filled landfills and toxic footprints. (due to incineration)
7. Health Problems due to accumulated plastic toxic gas can be reduced.
8. Minimising nuisance from leachate and odor.

Idea Title

Sustained Loop for Effective Plastic Waste Management and Community Benefit.

Idea in Explanation

By adding up a bit of psychological aspects, we are making the problem plastic waste management into a benefit for the whole society.

Our idea includes a system, a loop, in which the plastic waste are collected from the public in an economy.

When the plastic wastes are collected, a certain affordable amount is collected from the individuals.

During the time of collection of the plastic waste, based on the quality of the waste and the amount of the waste provided, a particular point is given to each user for their contribution. The points are given based on the quality whether the plastic materials are wet or dry or separated or cleaned thoroughly so that waste materials like organic food waste will be absent in those plastic waste provided.

From these plastic waste collected, materials like LDPE, HDPE, PP, PS will be processed in a processing center in that ULB itself and that will undergo chemical processes like pyrolysis, etc. Resulting in the formation

of valuable domestic fuels like LPG, CNG, petrol, diesel, etc. Where materials like PVC, those are not ideal for pyrolysis, are sent for recycling and thereby it will be also reused again. Later, a particular percentage of the amount of LPG, CNG or those domestic fuels obtained from those plastic waste by means of process like pyrolysis, etc are given back to the individuals based on the point they got or products usefull for the them are given to them.

The Process is done as follows (Theoretically sourced from sources like google, gemini, etc..)

Step -1 : feedstock determination

Standard grade fuels require feedstock to be fed in ratio to obtain it.

Target Primary Fuel	Ideal Feedstock Blend	Target Processing Temperature	Key Catalyst
High-Quality Diesel	70% HDPE + 30% LDPE	380°C -420°C	Mild Silica-Alumina (low cracking)
High-Octane Petrol	50% PP + 30% PS + 20% LDPE	450°C - 500°C	ZSM-5 Zeolite (acidic cracking)
LPG & CNG	60% PP + 40% PE (High Heat)	550°C - 650°C	H-ZSM-5 (high thermal/acid cracking)

Step-2 : Production Process

1.Pre-FeedstockTreatment

Contaminant Control

Shred plastic to 10–15 mm. Pass through a hot melter/extruder (200°C) under a nitrogen sweep. This melts the plastic, purges oxygen (preventing fires), and lets trapped moisture escape before entering the reactor.

2.Primary Continuous Pyrolysis

Polymer Depolymerisation

Feed molten plastic continuously into a stirred-tank reactor. Maintain an oxygen-free atmosphere heated to 420°C – 480°C. Solid plastic vaporizes into raw hydrocarbon gas, leaving behind ~5–8% dry carbon black residue (char).

3.In-Line Catalytic Vapor Cracking

Chain Length Reduction

Pass the raw vapours through a fixed-bed reactor filled with catalysts like ZSM-5 Zeolite catalyst at 400°C. This acts as molecular scissors: it breaks long, heavy waxes into liquid-range chains and light gases.

4. Fractional Distillation Column

Stream Separation

Direct cracked vapours into a multi-stage distillation column with distinct temperature cut-offs:

- Bottom (250°C – 350°C) Heavy Gas Oil / Diesel fraction
- Middle (150°C – 250°C): Kerosene / Jet Fuel fraction.
- Top (40°C – 150°C): Naphtha / Petrol fraction.
- Overhead Gas (Below 40°C): Non-condensable LPG/CNG gases.

5. Hydrotreating (The Commercial Quality Step)

Saturation & Purification

This is where raw oil is converted into certified fuel. Pump the condensed liquid streams into a Hydrotreater at 300°C under high pressure (30–50 bar) with hydrogen gas over a Nickel-Molybdenum (NiMo) catalyst. This:

- Saturates double bonds (eliminates unstable olefins that form engine gum).
- Removes trace chlorine, nitrogen, and oxygen.

6. Gas Cleanup, Scrubbing & Compression

LPG & CNG Finishing

Pass non-condensable overhead gases through an Amine Scrubber and Guard Beds (activated carbon/zinc oxide) to strip out acidic gases and trace chlorine.

CNG Stream : Compress to 200–250 bar for vehicle or pipeline use.

LPG Stream: Compress to 6–8 bar to liquefy into standard LPG cylinders.

Step-3 : Achievement of Standard Fuel grade.

To meet official fuel standards (such as **BS-VI**, **EN 590**, or **ASTM D975**), final liquid fuels must be blended with performance additives:

Commercial Diesel Finishing

- **Cetane Improver (2-EHN):** Boosts the cetane number to 51+ for fast ignition.
- **Cold Flow Improver (Anti-waxing additive):** Prevents wax crystallization during winter.
- **Lubricity Additive:** Protects modern high-pressure fuel pumps.

Commercial Petrol Finishing

- **Octane Booster (Aromatic blending or ETBE):** Raises Research Octane Number (RON) to 91–95.
- **Detergent Additives:** Keeps engine intake valves and fuel injectors clean.

LPG Safety Finishing

- **Mercaptan Dosing:** Raw synthetic LPG is completely odorless. You must inject trace amounts of **ethyl mercaptan** at 15–20 ppm so gas leaks can be detected by air/nose.

Why is this problem important to solve?

Addressing plastic waste management remains an urgent environmental issue, particularly in regions like Kerala where rapid urbanization and escalating consumption patterns have severely strained traditional disposal mechanisms. While the state government has pioneered commendable grassroots and policy initiatives—such as the Haritha Karma Sena, statewide bans on single-use plastics, and localized Green Protocols—these efforts encounter critical limitations that hinder their long-term efficiency. Inconsistent public cooperation, resistance to user fees, and a general lack of direct economic incentives for households to meticulously clean, dry, and segregate their plastic

waste. Overcoming these systemic gaps is of paramount relevance. While projects with reward-driven circular recycling loops is essential to eliminate open-air incineration hazards, curb illegal dumping, and secure a sustainable, pollution-free future for both society and the biosphere.

Advantages of the idea proposed

1. Plastic pollution can be reduced.
2. Idea acts as a strategy for gaining community acceptance.
3. Attitude of public can be changed for proper plastic waste management.
4. Increasing willingness to pay.
5. Proper Plastic waste management obtained with public support.
6. No more plastic filled landfills and toxic footprints. (due to incineration)
7. Health Problems due to accumulated plastic toxic gas can be reduced.
8. Minimising nuisance from leachate and odor.
9. Dependency for fuels for domestic use (LPG, oil, CNG) can be reduced at least to a small extent.
10. Economy develops.

Evidence for Problem Relevance











Stakeholder Survey Report: Sustained Loop for Effective Plastic Waste Management and Community Benefit.

Public Response

Sample Size: **10** Public Participants

Status: Initial **Concept Validation**

QUICK SUMMARY

Project Name: **Sustained Loop for Effective Plastic Waste Management and Community Benefit.**

People Surveyed: **10** participants

Main Takeaway: The idea is generally appreciated by the participants giving an overall **positive** response.

WHO PARTICIPATED?

- AGES **18 – 25** : **3** PEOPLE
- AGES **26 – 45** : **4** PEOPLE
- AGES **46+** : **3** PEOPLE

OBJECTIVES & PROJECT OVERVIEW

Concept Overview: A service in which plastic waste is collected from individuals for a particular affordable cost and analysed and user will get points based on the quantity and quality of the plastic like whether dry or wet ,separated or free from food stains. You will get maximum points if the plastic are dried,separated,free from food stains.Later,at the time of next collection LPG,CNG,diesel or petrol or useful products are given to the user based on the points.

Key Objectives:

- Analyse general **public interest** and initial impressions of POLYMERIT.
- Identify how much **acceptance** and **willingness** vary across age demographics.

What people liked?

- The proposal that they will get **usefull fuel/products** from their plastic waste.
- The way the idea **acts** to make others **aware/responsible of their own plastic waste**.

What confused people?

- The **scientific side** of the idea made the people confused.
- The **practical aspect** made the people hard to understand.

How they responded?

Age group	Response	Primary focus
18 - 25	Very interested	Focus on innovation and scientific aspects.
26 - 45	Interested	Focus on cost and practicality.
46+	Neutral	Focus on Trust Factors and simplicity.

EXECUTIVE SUMMARY

Project Name: **Sustained Loop for Effective Plastic Waste Management and Community Benefit.**

Survey Purpose: **Public Response & Concept Validation**

Sample Size: N = **10** participants

Overall Statement: This report summarizes initial public feedback on POLYMERIT gathered from a 10-person representative sample across three age brackets. Overall sentiment was **80% positive**, with high enthusiasm for the concept's potential.

-----**THE END**-----